Programme : **Diploma in Cloud Computing and Big Data**	
Course Code : **5273A**	Course Title: **Introduction to Microservices**
Semester : **5**	Credits: **4**
Course Category: **Programme Elective course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To provide an overview of Microservices Architecture
- To impart concepts of Monolithic Architecture and its drawbacks

Course Prerequisites:

Topic	Course Code	Course name	Semester
Cloud Computing Fundamentals	3273	Introduction to Cloud Computing	3
Operating System concepts	3271	Operating System	3
Database Management concepts	3272	Database System Concept	3
Advanced Operating System concepts	4279	Advanced Operating System	4
Web Application Development concepts	4277	Web Application Development Lab	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe microservices and monolithic architecture	13	Understanding
CO2	Discuss drawbacks of monolithic architecture and benefits of microservice architecture	16	Understanding
CO3	Explain how microservices interact with each other	14	Understanding
CO4	Illustrate the microservices based application with the focus on non functional requirements	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe microservices and monolithic architecture		
M1.01	Express overview of Monolithic architecture	3	Understanding
M1.02	Discuss Limitations of monolithic architecture	3	Understanding

<table>
<tr><td>M1.03</td><td>Explain Microservice architecture</td><td>3</td><td>Understanding</td></tr>
<tr><td>M1.04</td><td>Explore Features of microservice architecture</td><td>4</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Overview of Monolithic architecture : Definition, block diagram, example.
Limitations of monolithic architecture: Slow speed of development, High code coupling, Code ownership cannot be used, Testing becomes harder, Performance issues, The cost of infrastructure, Legacy technologies, lack of flexibility, Problems with deployment.
Microservice architecture: Definition, Architecture diagram, API Gateway, Service Discovery
Features of microservice architecture : Small focused, loosely Coupled, Language neutral, bounded Context.</td></tr>
<tr><td>CO2</td><td colspan="3">Discuss drawbacks of monolithic architecture and benefits of microservice architecture</td></tr>
<tr><td>M2.01</td><td>Compare Monolithic and Microservices architecture</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Discuss Advantages of microservice architecture</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Summarize Challenges of microservice architecture</td><td>4</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Discuss the benefits of Microservice architecture</td><td>4</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Content:
Comparison between Monolithic and Microservices : Basic, scale, database, Deployment, Tightly Coupled and Loosely coupled.
Benefits of Microservice architecture:
Technical Benefits: Replacing Microservices, Sustainable Software Development, Handling Legacy, Continuous Delivery, Scaling, Robustness, Free Technology Choice, Independence.
Organizational Benefits: Smaller Projects.
Benefits from Business perspective: Parallel Work.
Advantages of Microservices: Strong Modularization, Easy Replaceability, Sustainable development, Further Development of Legacy Applications, Time-to-Market, Independent Scaling, Free Choice of Technologies, Continuous Delivery
Challenges of microservice architecture: Relationships are hidden, Refactoring is difficult, Domain architecture is important, Running microservices is complex, Distributed systems are complex</td></tr>
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<tr><td>CO3</td><td colspan="3">Explain how microservices interact with each other</td></tr>
<tr><td>M3.01</td><td>Explain Communication challenges in microservices</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Discuss service to service communication in microservices</td><td>8</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Compare gRPC and REST</td><td>1</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Communication challenges in Microservices : Resiliency, Load balancing, Distributed tracing, Service versioning, TLS encryption and mutual TLS authentication.
Service to Service communication: Synchronous or Asynchronous Communication
Synchronous Communication : HTTP, HTTPS, GET, POST, PUT, DELETE, gRPC, RESTful APIs, features of REST, gRPC working and advantages
Asynchronous communication : Message Broker systems, Advanced Message Queuing Protocol- one-to-one(queue) mode or one-to-many (topic) mode.
gRPC and REST : compare based on maintenance, performance and compatibility</td></tr>
<tr><td>CO4</td><td colspan="3">Illustrate the microservices based application with the focus on non functional requirements</td></tr>
<tr><td>M4.01</td><td>Discuss non functional requirements of microservices</td><td>5</td><td>Understanding</td></tr>
<tr><td>M4.02</td><td>Summarize Scalability in microservices</td><td>5</td><td>Understanding</td></tr>
<tr><td>M4.03</td><td>Outline the Security structure in a microservice</td><td>5</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – II</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Non functional requirements of microservices: Security, Maintainability, Scalability, Performance, Reusability, Reliability, Modularity, Availability and Portability.
Scalability: Horizontal scalability, Vertical scalability, Scaling, Microservices and Load Balancing, Dynamic Scaling, Advantages of Scaling
Security: Security and Microservices Authentication and Authorization- OAuth2-JSON Web Token (JWT)</td></tr>
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Text / Reference

T/R	Book Title/Author
T1	“Microservices- Flexible Software Architecture”, Eberhard Wolff, Addison-Wesley
R1	"Monolith to Microservices: 7Evolutionary Patterns to Transform Your Monolith", Sam Newman , O'Reilly

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com/difference-between-monolithic-and-microservices-architecture
2	https://www.youtube.com/watch?v=tuJqH3AV0e8
3	https://www.tutorialspoint.com/microservice_architecture/microservice_architecture_quick_guide.htm
4	https://medium.com/design-microservices-architecture-with-patterns/microservices-communications-f319f8d76b71
5	https://learn.microsoft.com/en-us/azure/architecture/microservices/design/interservice-communication
6	springboottutorial.com/non-functional-requirements-in-microservices-introduction-to-performance